

Moon Phases Explorer

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01

Abstract

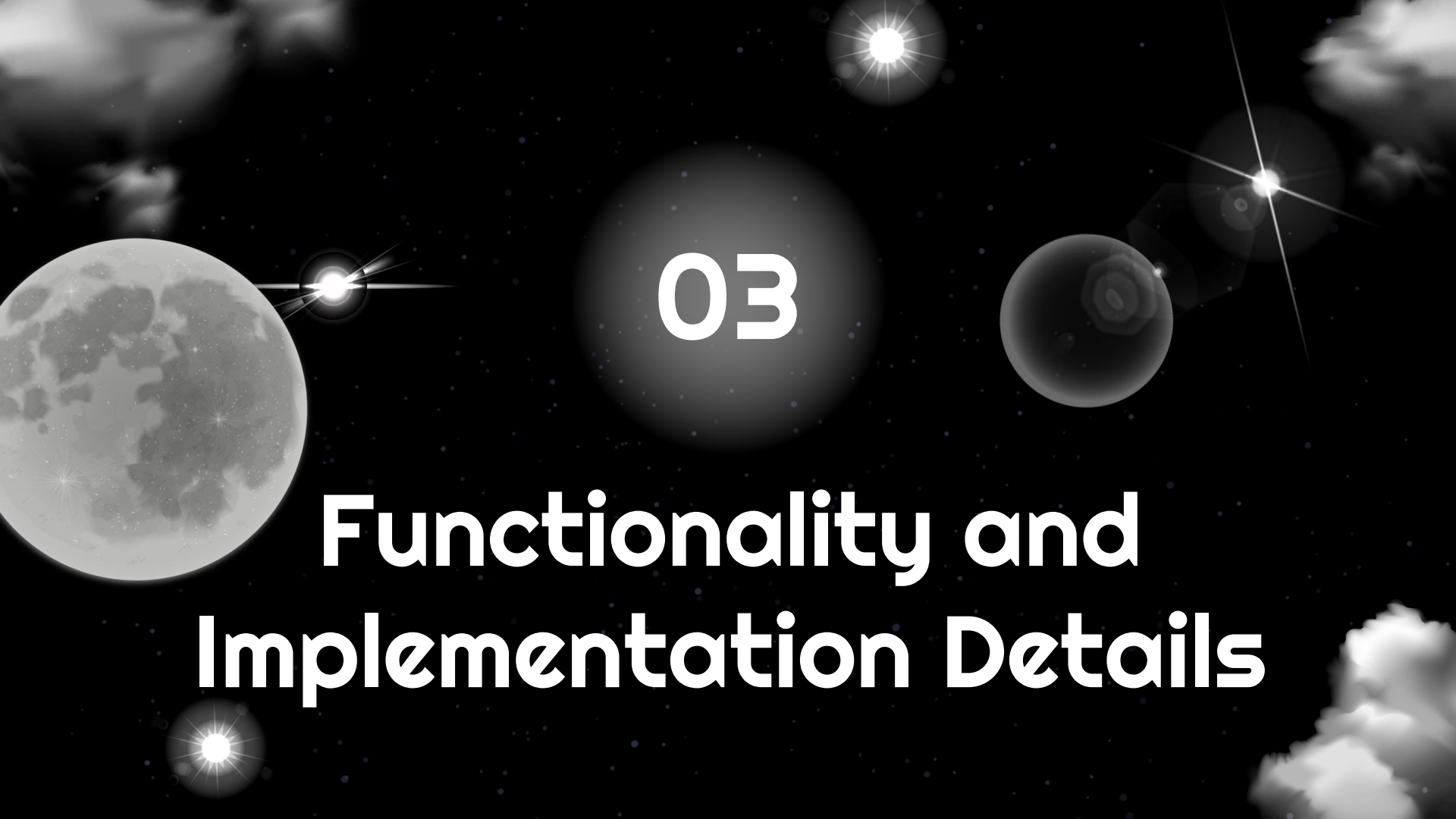
Abstract

Moon Phases Explorer is an interactive computer graphics project designed to visually simulate the movement of the Moon around the Earth and demonstrate the main phases of the Moon. The project uses OpenGL and GLUT to create a 3D space environment that includes the Earth, the Moon, orbit path, stars, lighting effects, and phase markers. The main goal of the project is to help users understand the relationship between the Moon's position and its visible phases in an interactive and visual way. The contribution of this project is combining educational content with graphical interaction, allowing the user to select moon phases, pause the animation, zoom in and out, and rotate the camera to explore the scene from different angles.


02
Introduction

Introduction

The Moon Phases Explorer project presents an interactive 3D simulation to help users understand the main phases of the Moon. It shows the Earth at the center and the Moon moving around it, with the eight major phases marked on the orbit. Users can interact with the simulation by selecting a phase, viewing its description, pausing or resuming the animation, rotating the camera, zooming, and resetting the view. The project focuses on educational visualization using simple 3D graphics, animation, lighting, textures, and user interaction rather than scientific accuracy.


03
Functionality and
Implementation Details

Functionality and Implementation Details

01

The project was developed using C++ with OpenGL, GLU, GLUT, and GLEW.

02

The Earth is displayed as a rotating textured 3D sphere.

03

The simulation creates a 3D space environment with a dark background, stars, lighting, and textures.

Functionality and Implementation Details

04

The Moon is displayed as a textured 3D sphere moving around the Earth in a circular orbit.

05

The system includes eight moon phase markers around the orbit, and the selected phase is highlighted.

06

Users can interact with the simulation using the mouse and keyboard to select phases, pause, rotate, zoom, reset, or exit.

Functionality and Implementation Details

07

When a phase is selected, the system displays its name, short description, and a small visual preview.

04

Additional Libraries Used

Additional Libraries Used

OpenGL

Used for rendering 2D and 3D graphics.

GLU

Used for drawing spheres and setting the perspective projection.

GLUT / FreeGLUT

Used for window creation, keyboard input, mouse input, and the main loop.

GLEW

Used to initialize OpenGL extensions on non-Apple systems.

cmath

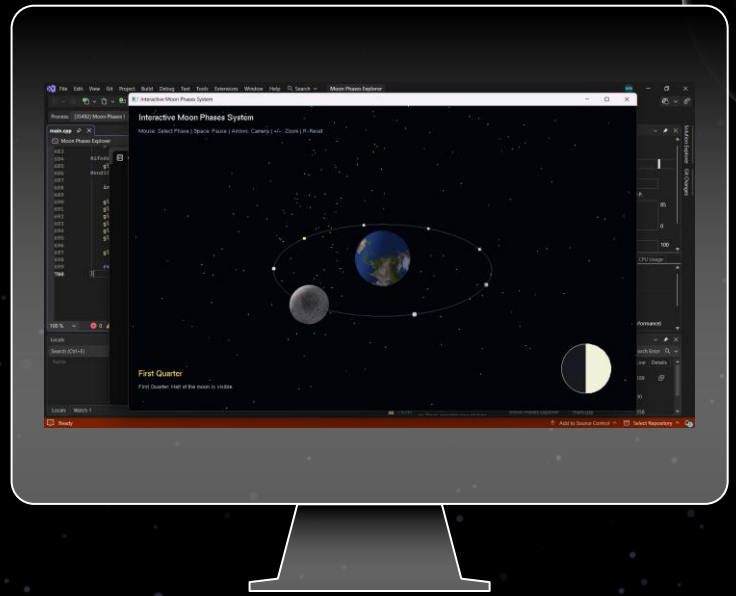
Used for mathematical calculations such as sine, cosine, and square root.

cstdio /cstdlib /iostream

Used for file handling, console output, and program control.

05

Result

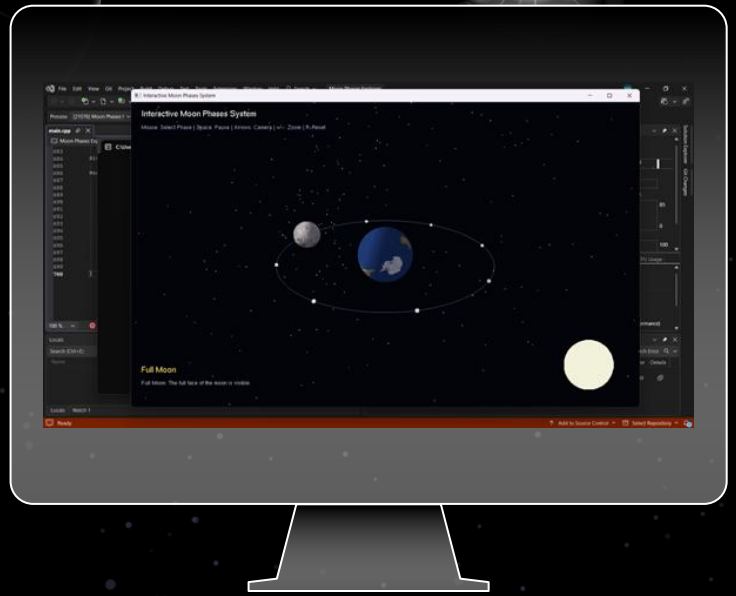


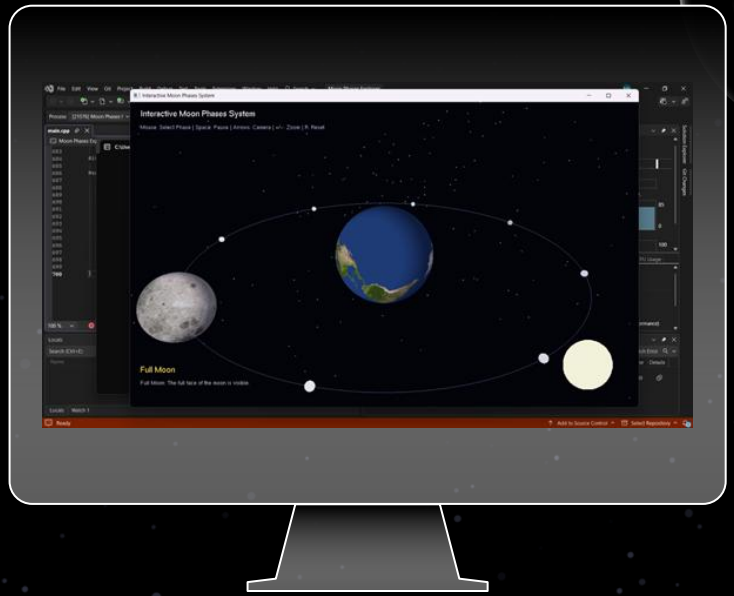
Main Simulation Window

The main window demonstrates the general structure of the project. It shows the Earth, the Moon, the orbit path, and the space environment. The camera view allows the user to observe the system from a clear angle.

lected Moon Phase Information

This feature improves the learning experience by giving the user direct information about the selected phase. The small moon phase preview also helps the user understand how the Moon appears during that phase.





Camera Rotation and Zoom

The camera controls allow the user to explore the scene from different views. This makes the simulation more flexible and helps the user understand the 3D arrangement of the Earth, Moon, and orbit.

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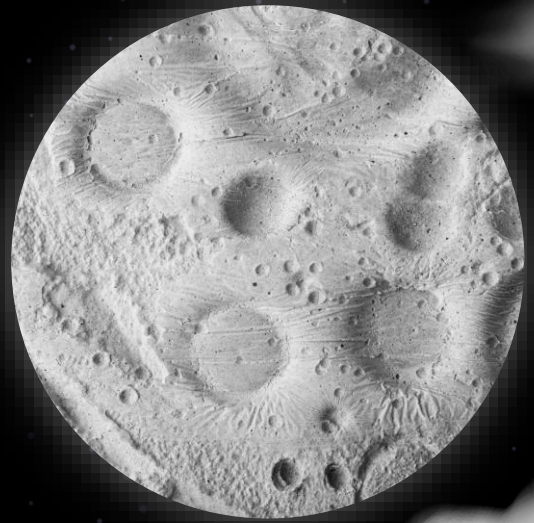
Conclusion and Future Work

Conclusion and Future Work

Moon Phases Explorer successfully presents an interactive 3D simulation that explains the main moon phases in a simple and visual way. The project combines animation, lighting, textures, and user interaction to make learning easier.

The project also demonstrates important computer graphics concepts such as 3D transformations, texture mapping, lighting, camera control, and event handling.

In the future, the project could be improved by adding more realistic calculations, a visible Sun, better textures, sound effects, phase labels, a lunar cycle timeline, and a menu to control animation speed or switch between modes.




Thank You